Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Ramalingam SS, Vansteenkiste J, Planchard D, et al. Overall survival with osimertinib in untreated, *EGFR*-mutated advanced NSCLC. N Engl J Med 2020;382:41-50. DOI: 10.1056/NEJMoa1913662

(PDF last updated on February 4, 2020.)

Overall Survival with Osimertinib in Previously Untreated EGFR-Mutated Advanced Non–Small-Cell Lung Cancer

SUPPLEMENTARY APPENDIX

This appendix is provided by the authors to give readers additional information about their work.

Supplement to: Ramalingam SS, Vansteenkiste J et al. Overall Survival with Osimertinib in Previously Untreated EGFR-Mutated Advanced Non–Small-Cell Lung Cancer

Contents

LIST OF FLAURA INVESTIGATORS 3

SUPPLEMENTARY METHODS 5

Study oversight 5

Trial design and treatment 5

Statistical methods 5

Global interaction test 5

SUPPLEMENTARY RESULTS 7

Efficacy 7

Safety 7

Adverse events leading to permanent discontinuation, dose interruption or dose reduction 8

Cardiac effects 8

SUPPLEMENTARY FIGURES 10

Figure S1. Patient disposition 10

Figure S2. Probability of overall survival by race and *EGFR* mutation subtype 11

Figure S3. Probability of no first subsequent therapy or death and no second subsequent therapy or death 13

SUPPLEMENTARY TABLES 15

Table S1. Global interaction of all predefined subgroups for overall survival (Panel A). Interaction tests in separate simplified models for race and *EGFR* mutation subtype for overall survival (Panel B). 15

Table S2. *EGFR* mutation subtype by race 16

Table S3. First and second subsequent therapies* 17

Table S4. Subsequent therapies by race (Panel A) and by EGFR mutation subtype (Panel B) 19

Table S5. Adverse events possibly causally-related to study treatment (investigator assessed) reported in at least 10% of the patients treated in either treatment arm 25

Table S6. Adverse events of grade 3 or higher by preferred term 26

Table S7. Serious adverse events by preferred term 32

REFERENCES 37

LIST OF FLAURA INVESTIGATORS

Australia: Michael Boyer, Chee Lee, Brett Hughes, Kenneth O'Byrne, Peter Briggs, Michael Milward, Thomas John

Belgium: Ingel Demedts, Johan Vansteenkiste, Frédérique Bustin

Brazil: Carlos Henrique Barrios

Bulgaria: Constanta Timcheva

Canada: Charles Butts, Glenwood Goss, Rosalyn Juergens, Natasha Leighl, Susanna Cheng, Ronald Burkes, Parneet Cheema

China: Caicun Zhou, Helong Zhang, Yongqian Shu, Ying Cheng, Qing Zhou, Wei Li, Guosheng Feng, Yong He, Buhai Wang, Hongjun Gao, Kejun Nan, Xiangdong Zhou, Yun Fan, Jian an Hunag, Yi Hu, Yun-Peng Liu, Chunling Liu, Yiping Zhang, Bo Zhu

Czech Republic: Tomas Bartek

France: Isabelle Monnet, Claude El Kouri, Maurice Perol, Henri Berard, David Planchard, Jeannick Madelaine

Germany: Joachim von Pawel, Christian Grohe, Sabine Bohnet, Helge Bischoff, Christian Meyer zum Büschenfelde, Anja Rückert

Hungary: Lazlo Urban, Zsolt Papi-Szekely, Eszter Csánky, Csaba Böcskei, Sándor Tehenes

Israel: Maya Gottfried, Jair Bar, Nir Peled, Mirjana Wollner

Italy: Alessandro Bertolini, Antonio Ardizzoia, Caterina Accettura, Michele Mitella, Rodolfo Passalacqua, Silvia Novello, Fausto Roila, Marcello Tiseo

Japan: Akimasa Sekine, Fumio Imamura, Yuichiro Ohe, Naoyuki Nogami, Toshiaki Takahashi, Takayasu Kurata, Kazuo Kasahara, Isamu Okamoto, Kiyotaka Yoh, Shunichi Sugawara, Nobuaki Kobayashi, Tsuneo Shimokawa, Chiyuki Okuda, Tatsuo Fukuhara, Masafumi Sata, Kazuhiko Nagakawa, Shinji Atagi, Yoshiro Nakahara

Republic of Korea: Byoung Chul Cho, Jong-Seok Lee, Kye Young Lee, Sang Won Shin, Ki Hyeong Lee, Eun Kyung Cho, Jin-Hyoung Kang

Malaysia: Soon Hin How, Pei Jye Voon, Yong Kek Pang

Philippines: Jemela Anne Osorio-Sanchez, Guia Elena Imelda Ladrera

Poland: Piotr Serwatowski, Aleksandra Szczesna, Katarzyna Zajda, Rodryg Ramlau, Dariusz Sawka

Portugal: Maria Encarnação Teixeira, Teresa Almodovar, Marta Soares, Amélia Almeida, Ana Barroso

Romania: Michael Schenker, Alexandru Calin Girgorescu, Polixenia Iorga

Russian Federation: Sergey Orlov, Nina Karaseva, Vladimir Moiseenko

Spain: Rosario Garcia Campelo, Maria Dolores Isla Casado, Manuel Cobo Dols, Ernest Nadal, Marc Campayo Guillaumes, Jose Luis González Larriba, Margarita Majem Tarruella, Sergio Vázquez Estévez, José Fuentes Pradera¸ Diego Márquez Medina

Sweden: Anders Vikstrom

Switzerland: Oliver Gautschi, Christian Britschgi, Andreas Mueller

Taiwan: Ying-Huang Tsai, Ming-Fang Wu, Cheng-Ta Yang, Meng Chih Lin, Chao-Hsun Chen, Wu-Chou Su

Thailand: Busyamas Chewaskulong, Thanyanan Reungwetwattana, Vichien Srimuninnimit, Arunee Dechaphunkul, Sudsawat Laohavinij, Virote Sriuranpong

Turkey: Ramazan Yildiz

Ukraine: Igor Bondarenko, Ihor Vynnychenko, Hryhoriy Adamchuk, Yaroslav Shparyk

United Kingdom: Siow Ming Lee, Allan Price, Yvonne Summers, Riyaz Shah

Unites States of America: Konstantin Dragnev, Ajit Maniam, Steven McCune, Jimmy Ruiz, Suresh Ramalingam, Pasi Jänne, Trevor Feinstein, John Hamm, Ian Anderson, Farrah Khan, Ralph Boccia, Jhanelle Gray

Vietnam: Ngoc Tran, Nhung Nguyen

SUPPLEMENTARY METHODS

Study oversight

All authors signed a confidentiality agreement with the sponsor. An agreement was in place between the study sponsor and the authors, which established the authors' rights to publish the study and access the data. Responsibility for opinions, conclusion, and interpretation of the data lies with the authors.

Trial design and treatment

A protocol amendment on April 13, 2015, allowed patients randomly assigned to a standard EGFR-TKI to cross over to open-label osimertinib after confirmation of disease progression (during trial treatment or within 28 days of trial treatment cessation with no intervening therapy).

Statistical methods

Overall survival was analyzed using a log-rank test stratified by race (Asian vs non-Asian) and mutation type (Ex19del vs. L858R) for generation of the P value, and the Breslow approach was used for handling ties. The hazard ratio and confidence intervals were obtained directly from the U and V statistics.[1,2] The minimum critical hazard ratio for overall survival was defined as 0.801, with a two-sided alpha of 0.0495 with 318 death events. The covariates in the statistical modelling were based on the values entered into the interactive voice recognition system at randomization, even if it was subsequently discovered that these values were incorrect.

Overall survival was analyzed for predefined subgroups. Each subgroup required at least 20 overall survival events per subgroup level for the data to be included in the analysis. The analysis was performed using a Cox proportional hazards model, including treatment, subgroup, and a treatment-by-subgroup interaction term for each subgroup to provide the hazard ratio and 95% confidence interval. This was then presented on a forest plot.

Global interaction test

A global interaction test was used to assess whether there was a consistent treatment effect across the predefined subgroups. This test was performed using the overall population. The fit of the Cox proportional hazards model, including treatment, covariates for race (Asian vs. non-Asian) and mutation type (Ex19del vs. L858R), and all covariate by treatment interaction terms, was compared with a model that excluded the interaction terms. This comparison was assessed at the two-sided 10% significance level. If the Cox proportional hazards model was not significantly improved it would be concluded that the treatment effect is consistent across subgroups. If the global interaction test was statistically significant, a step-wise backwards

selection would be performed on the saturated model to attempt to determine the cause and type of interaction. Throughout this process, all the main effects remain in the model regardless of whether the corresponding interaction term was present, with the least significant interactions excluded one-by-one. Any newly significant interactions were re-included until a final model was reached where all included interactions were significant and all excluded interactions were non-significant. This enables identification of the factors that potentially alter the treatment effect. Any quantitative interactions identified using this procedure were then tested to rule out any qualitative interaction using the approach of Gail and Simon.[3]

SUPPLEMENTARY RESULTS

Efficacy

The median overall survival was 38.6 months (95% confidence interval [CI], 34.5 to 41.8) in the osimertinib group and 31.8 months (95% CI, 26.6 to 36.0) in the comparator group. The hazard ratio for death was 0.799 (95.05% CI 0.641 to 0.997), P=0.0462, which is below the overall survival critical hazard ratio of 0.801. A global interaction test was conducted to assess whether a consistent treatment effect was present across all predefined subgroups. A statistical difference in the magnitude of treatment effect between Asian and non-Asian patients was observed at the 10% significance threshold (P=0.0127) (Table S1).The Kaplan-Meier event curve in the Asian subset shows separation from time of randomization until approximately 3 years (Fig. S2A and 2C).

There were 70 patients in the comparator arm who crossed over to osimertinib within the study and an additional 15 patients received it outside the study. Of those patients in the comparator EGFR-TKI arm, more patients with an Ex19del than L858R EGFR mutation subtype crossed over to receive osimertinib: 53 of 174 patients (30%) vs 17 of 103 patients (17%), respectively. *EGFR* mutation subtype by race is reported in Table S2. Of those Asian and non-Asian patients in the comparator EGFR-TKI arm, 39 of 173 patients (23%) and 31 of 104 patients (30%), respectively, crossed over to receive osimertinib within the study.

Osimertinib extended the time to first and second subsequent therapies or death versus the comparator EGFR-TKI arm (HR 0.48; 95% CI, 0.39 to 0.58; and HR 0.69; 95% CI, 0.56 to 0.84, respectively; Fig. S3).

Safety

The most commonly reported adverse events due to any cause (treatment-related or not) were diarrhea (60% in the osimertinib arm and 58% in the comparator EGFR-TKI arm), rashes and acnes (grouped term; 59% and 79%, respectively), and nail effects (grouped term; 39% and 34%, respectively) (Table 2). Adverse events that were considered by the investigator to be causally related were reported by 91% and 92% of patients in the osimertinib and comparator EGFR-TKI arms, respectively (Table S5). Overall, serious adverse events were reported in 74 patients (27%) in the osimertinib arm and 76 patients (27%) in the comparator EGFR-TKI arm (Table S7).
Fatal adverse events occurred in nine patients (3%) in the osimertinib arm. These included: pneumonia (n=2), sepsis and renal failure (n=1), respiratory tract infection (n=1), cerebral infarction (n=1), cerebral ischemia (n=1), myocardial infarction (n=1), pulmonary embolism (n=1), intestinal ischemia (n=1). In the comparator EGFR-TKI arm, fatal events occurred in

10 patients (4%): pneumonia (n=1), sepsis (n=2), cognitive disorder and pneumonia (n=1), endocarditis (n=1), pericarditis (n=1), peripheral artery occlusion (n=1), hemoptysis (n=1), diarrhea, respiratory failure, circulatory collapse and gastrointestinal hemorrhage (n=1), unknown cause of death (n=1). All fatal events in the osimertinib group were considered unrelated to study treatment by the Investigators. Two fatal events in the comparator group were considered possibly causally related to treatment: pericarditis (n=1) and diarrhea, circulatory collapse and gastrointestinal hemorrhage (n=1).

Adverse events leading to permanent discontinuation, dose interruption or dose reduction

Permanent discontinuation attributed to adverse events occurred less frequently in the osimertinib arm compared with the comparator EGFR-TKI arm (41 patients [15%] and 50 patients [18%], respectively). The most common events leading to permanent discontinuation in the osimertinib arm were interstitial lung disease (n=6), pneumonitis (n=5) and QT prolongation (n=4), while in the comparator EGFR-TKI arm they were alanine and aspartate aminotransferase increases (n=12 and n=8, respectively). The frequency of adverse events leading to dose interruptions (29% in the osimertinib arm and 26% in the comparator EGFR-TKI arm) and dose reductions (4% in the osimertinib arm and 6% in the comparator EGFR-TKI arm) were similar in the two arms. The most frequently experienced adverse events leading to dose interruptions in the osimertinib arm were diarrhea (n=9), decreased appetite (n=9), QT prolongation (n=8), and pneumonia (n=5); in the comparator EGFR-TKI arm they were increased alanine aminotransferase (n=18), increased aspartate aminotransferase (n=12), QT prolongation (n=6), and decreased appetite (n=6). Dose reductions were primarily driven by QT prolongation (n=5) in the osimertinib arm and by skin disorders (n=10), in the comparator EGFR-TKI arm.

Cardiac effects

Adverse events of ejection fraction decrease were reported in 13 patients (5%) in the osimertinib arm vs. 5 patients (2%) in the comparator EGFR-TKI arm, with no symptoms reported. Of those patients with a baseline and at least one follow-up left ventricular ejection fraction assessment, the ejection fraction decreased at least 10 percentage points to a value of less than 50% in 12 of 267 patients (4%) in the osimertinib arm and 4 of 271 patients (1%) in the standard EGFR-TKI arm.

Electrocardiogram QT prolongation events were reported in 28 patients (10%) in the osimertinib arm vs 12 patients (4%) in the comparator EGFR-TKI arm). Changes in QT interval of grade 3 or higher were reported in 4 patients (1%) and 2 patients (1%) in the osimertinib and comparator EGFR-TKI arms, respectively.

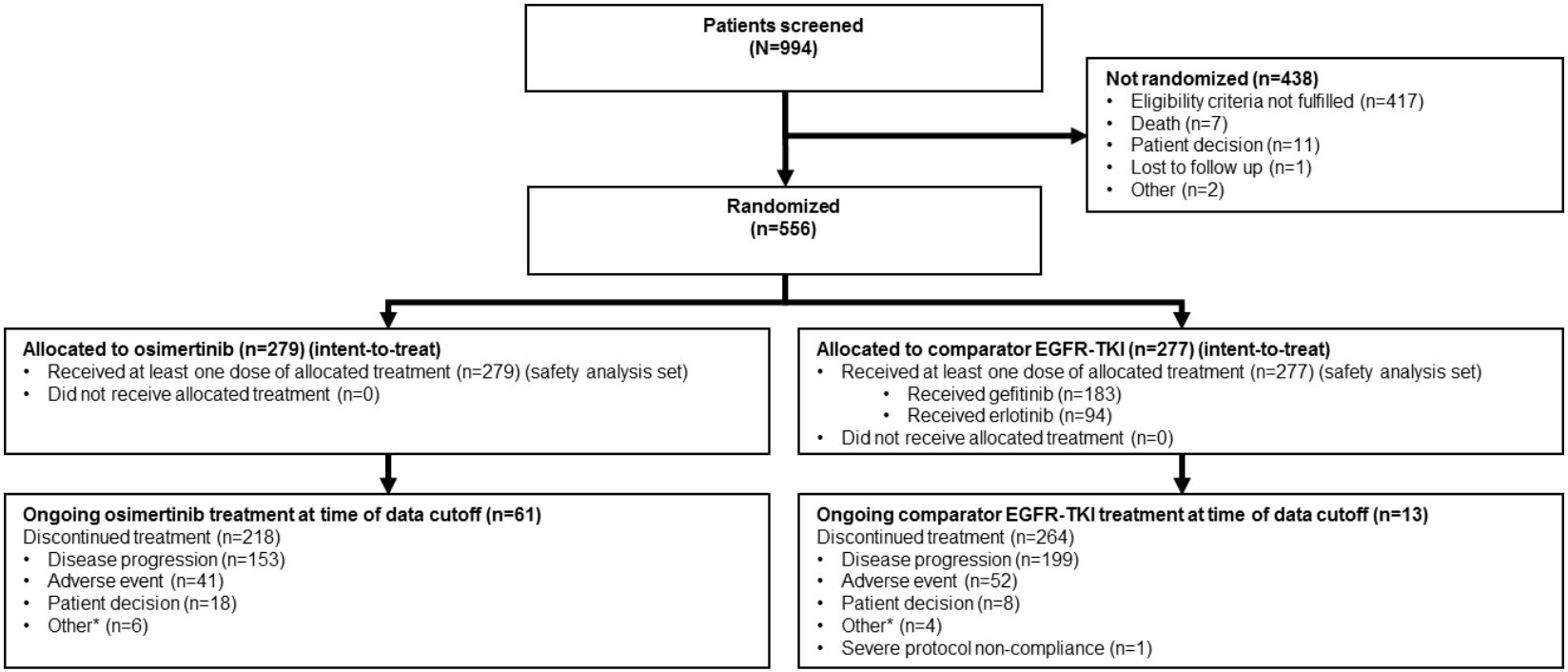
In the osimertinib arm, one patient discontinued study treatment after two cycles of treatment due to exacerbation of chronic heart failure (grade 3) and QT prolongation event (grade 2), both considered to be possibly related to treatment; this patient had a medical history of hypertension, chronic heart failure, valvular heart disease, cerebral infarction, hyperlipidemia and atrial fibrillation. Two patients in the osimertinib arm discontinued treatment due to adverse events of myocardial infarction; one was considered unrelated to treatment (grade 5) and the second was considered possibly related to treatment (grade 3). One patient discontinued due to an adverse event of conduction disorder (grade 2) possibly related to treatment with no symptoms reported. One report of tachyarrhythmia (grade 2) was reported, that was considered unrelated to treatment, following an adverse event of bilateral pulmonary thromboembolism. Four patients discontinued treatment due to an adverse event of electrocardiogram QT prolongation (grade 2, n=3; grade 3, n=1), all four patients had no symptoms; the patient with QT prolongation of grade 3 had a QTcF of 486 at the time of discontinuation.

In the comparator EGFR-TKI arm there was one report of treatment discontinuation due to ejection fraction decrease (51% down from 80% at baseline; grade 3), which was considered possibly related to treatment with no symptoms reported. One patient reported an adverse event of pericardial effusion (grade 2) considered possibly related to treatment, and one patient discontinued due to a QT prolongation event (grade 1) considered possibly related to treatment with no reported symptoms.

Of those patients who crossed over to osimertinib from the comparator EGFR-TKI arm there was one patient who discontinued treatment due to heart failure (grade 4) that was considered unrelated to treatment, and one patient discontinued due to QT prolongation (grade 1) that was considered possibly related to treatment with no reported symptoms.

SUPPLEMENTARY FIGURES

Figure S1. Patient disposition



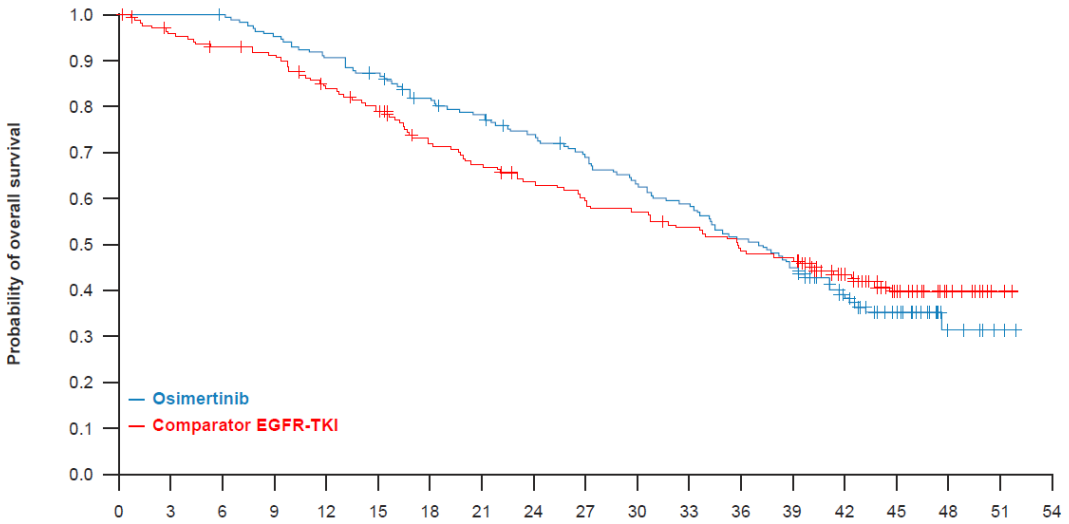
*Any reason not specifically recorded; for example, subject died.

EGFR denotes epidermal growth factor receptor, TKI tyrosine kinase inhibitor.

Figure S2. Probability of overall survival by race and *EGFR* mutation subtype

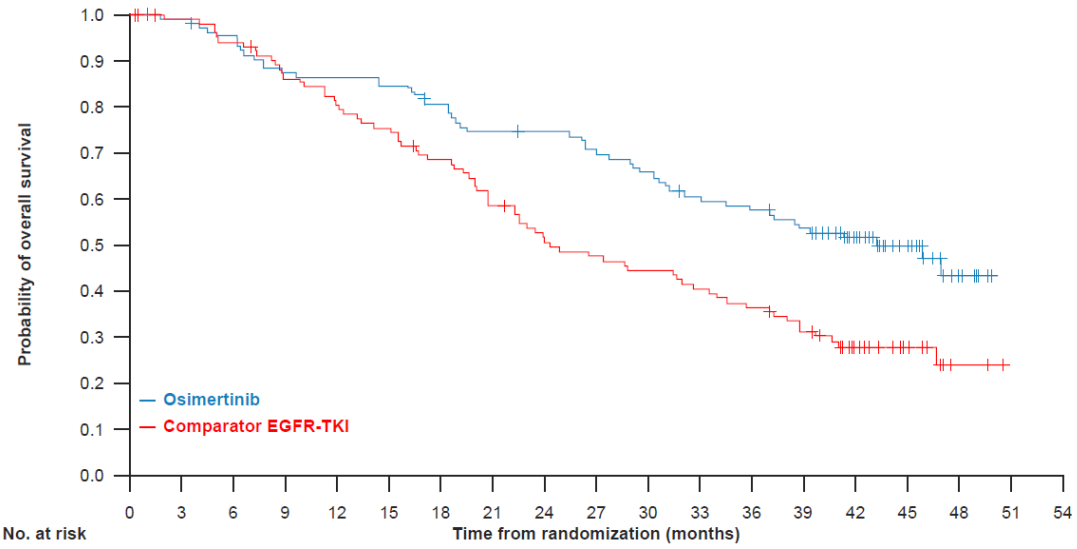
Shown are Kaplan-Meier estimates for probability of overall survival in Asian patients (Panel A), non-Asian patients (Panel B), in patients with an *EGFR* Ex19del mutation (Panel C), and in patients with an *EGFR* L858R mutation (Panel D). Censored data are indicated by tick marks. Data for patients who had not died at the time of the analysis were censored based on the last recorded date on which the patient was known to be alive.

A



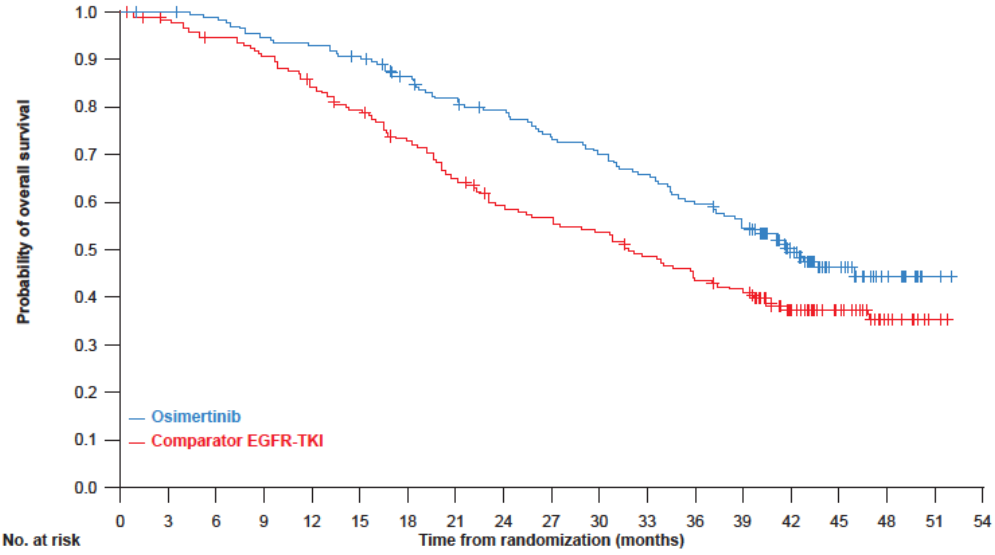
No. at risk										
Osimertinib	174	172	156	135	118	100	81	48	8	0
Comparator EGFR-TKI	173	156	137	113	97	87	74	52	15	0

B



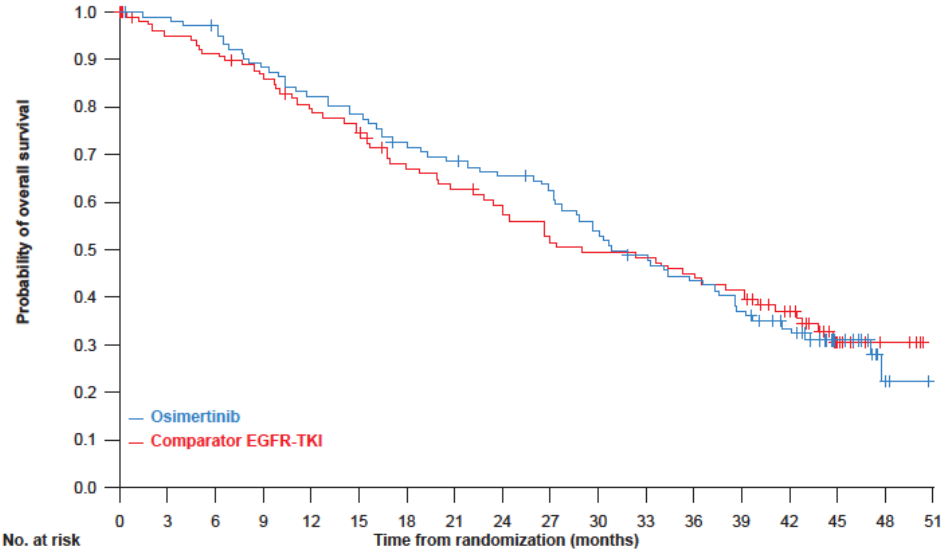
No. at risk										
Osimertinib	105	98	89	82	75	66	57	38	9	0
Comparator EGFR-TKI	104	96	82	69	51	44	36	20	2	0

C



No. at risk																		
Osimertinib	175	174	171	164	161	156	144	135	129	120	114	107	97	88	58	33	13	2	0
Comparator EGFR-TKI	174	168	161	154	142	133	120	107	95	91	86	77	69	64	43	29	12	2	0

D



No. at risk																		
Osimertinib	104	102	99	90	84	80	73	69	64	60	52	46	41	35	28	17	4	0
Comparator EGFR-TKI	103	95	91	85	77	72	62	58	53	47	45	44	41	37	29	11	5	0

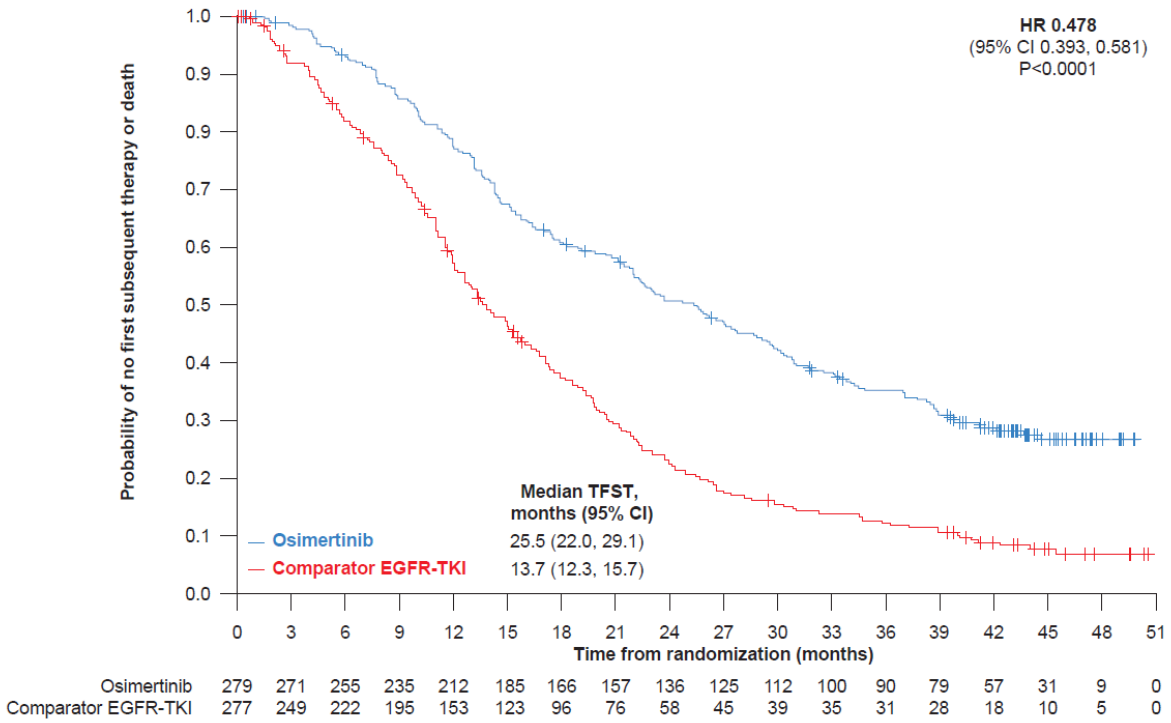
Figure S3. Probability of no first subsequent therapy or death and no second subsequent therapy or death

Shown are Kaplan-Meier estimates of time to first subsequent therapy (Panel A) and second subsequent therapy (Panel B).

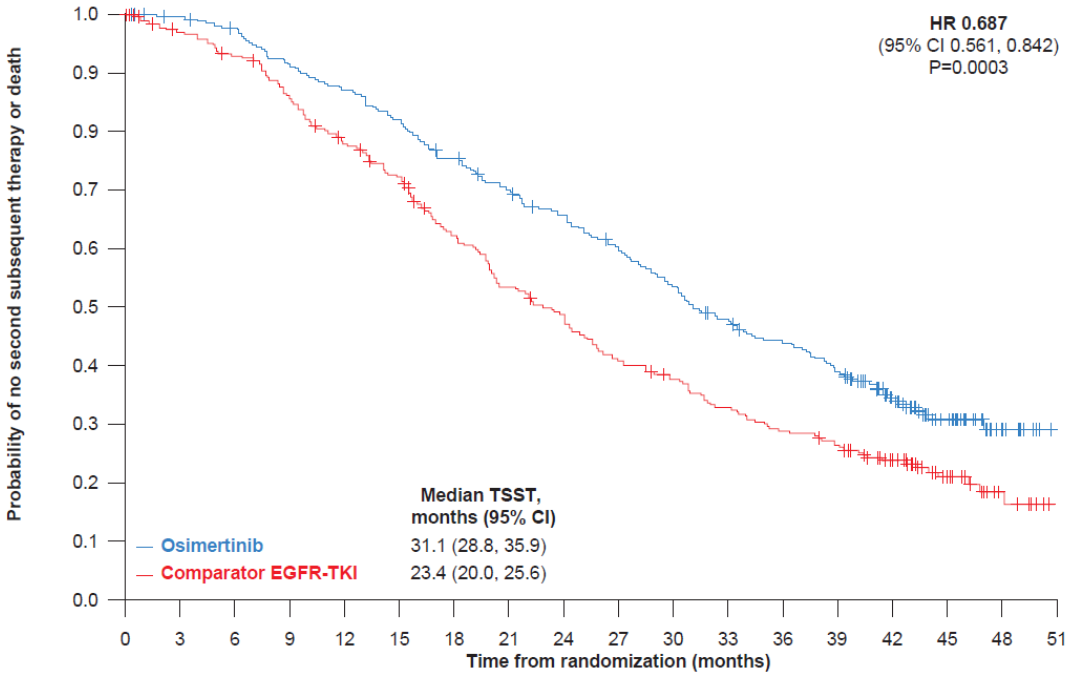
Censored data are indicated by tick marks. Data for patients who had not died at the time of the analysis were censored based on the last recorded date on which the patient was known to be alive.

TFST, time to first subsequent therapy or death; TSST, time to second subsequent therapy or death

A



B



Osimertinib	279	274	267	250	238	224	205	189	175	159	142	125	113	100	68	37	12	0
Comparator EGFR-TKI	277	263	251	230	207	189	159	137	123	104	94	82	72	65	46	25	9	0

SUPPLEMENTARY TABLES

Table S1. Global interaction of all predefined subgroups for overall survival (Panel A). Interaction tests in separate simplified models for race and *EGFR* mutation subtype for overall survival (Panel B).

A

	P-value of quantitative interaction test*	If significant, quantitative or qualitative difference observed	P-value of qualitative interaction test†
Global interaction test (all subgroups)	0.0815	Quantitative	
Race (Asian vs non-Asian)	0.0127	Quantitative	0.5000

B

	P-value of quantitative interaction test‡	If significant, quantitative or qualitative difference observed	P-value of qualitative interaction test†
Race (Asian vs non-Asian)	0.0087	Quantitative	0.5000
***EGFR* mutation type (Ex19del vs L858R)**	0.0930	Quantitative	0.5000

The analysis was performed using a Cox proportional hazards model including treatment, subgroup and a treatment-by-subgroup interaction term.

*The P-value for the Global interaction test represents the significance of there being a quantitative interaction in the model. P-values for covariates represent the significance of that treatment-covariate interaction term remaining in the final model based on a stepwise backwards selection. Significant covariates will be identified as the causes of a significant Global Interaction. Stepwise backwards selection was used to determine the cause of interaction.

†Test statistics presented have been compared to a critical value of 1.64 to determine the significance of a qualitative interaction. A test statistic P-value <0.1 suggests a qualitative interaction is present.

‡P-values for covariates represent the significance of that treatment-covariate interaction term remaining in the final model based on a stepwise backwards selection.

Table S2. *EGFR* mutation subtype by race

Race	*EGFR* mutation subtype	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)	Total (n=556)
		Number of patients (%)		
Asian	**Ex19del**	102 (37)	102 (37)	204 (37)
	L858R	72 (26)	71 (26)	143 (26)
Non-Asian	**Ex19del**	73 (26)	72 (26)	145 (26)
	L858R	32 (11)	32 (12)	64 (12)

EGFR, epidermal growth factor receptor; Ex19del, Exon 19 deletion mutations; TKI, tyrosine kinase inhibitor.

Table S3. First and second subsequent therapies*

Subsequent therapy	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
Number of patients		
Number of regimens		
1	68	96
2	35	47
3	12	20
4	14	11
≥5	4	6
First subsequent therapies		
Received a first subsequent therapy	133	180
Cytotoxic chemotherapy	90	39
Platinum-containing chemotherapy	84	36
EGFR-TKI†	38	49
Other TKI‡	2	3
Osimertinib	1	85
Immunotherapy	4	4
VEGF inhibitor§	10	4
Other¶	2	2
Second subsequent therapies		
Received a second subsequent therapy	72	92
Cytotoxic chemotherapy	38	56
Platinum-containing chemotherapy	15	42
EGFR-TKI‖	25	25
Immunotherapy	6	5
VEGF inhibitor**	11	11
Other	1	6

*The first post-treatment anticancer therapy was the first treatment started on or after the last dose date of randomized trial treatment. The second post-treatment anticancer therapy was the second treatment started on or after the last dose date of randomized trial treatment. Patients with no post-treatment anticancer therapy had discontinued randomized trial treatment without starting any other post-treatment anticancer therapy. Patients may be counted in multiple rows if they received more than one anticancer therapy or a combination therapy that contained drug substances from multiple classifications.

†Excluding osimertinib; including one patient in the osimertinib arm and three patients in the comparator EGFR-TKI arm who received an EGFR-TKI plus cytotoxic chemotherapy, and one patient in the osimertinib arm and two patients in the comparator arm who received an EGFR-TKI plus VEGF inhibitor.

‡One patient in the comparator EGFR-TKI arm received cytotoxic chemotherapy plus other TKI, and two patients in the osimertinib arm and one patient in the comparator arm received an EGFR-TKI plus other TKI.
§Ten patients in the osimertinib arm and four patients in the comparator EGFR-TKI arm received a VEGF inhibitor plus cytotoxic chemotherapy.
¶Six patients (osimertinib n=2, comparator EGFR-TKI n=4) received subsequent treatment of an investigational drug as a monotherapy.
‖One patient in the osimertinib arm and two patients in the comparator EGFR-TKI arm received an EGFR-TKI plus cytotoxic chemotherapy, and three patients in each arm received an EGFR-TKI plus VEGF inhibitor.
**Seven patients in the osimertinib arm and six patients in the comparator EGFR-TKI arm received a VEGF inhibitor plus cytotoxic chemotherapy, and one patient in the comparator EGFR-TKI arm received an EGFR-TKI plus cytotoxic chemotherapy and a VEGF inhibitor.
EGFR, epidermal growth factor receptor; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor

Table S4. Subsequent therapies by race (Panel A) and by EGFR mutation subtype (Panel B)

A

Subsequent therapy	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
	number of patients	
Asian	**n=174**	**n=173**
Received any subsequent therapy	83	105
Number of regimens		
1	39	55
2	23	25
3	7	14
4	12	6
≥5	2	5
Cytotoxic chemotherapy	64	58
Number of regimens		
1	45	42
2	12	11
3	6	3
4	0	2
≥5	1	0
Platinum-containing chemotherapy	60	52
EGFR-TKI	49	82
Number of regimens		
1	39	61
2	6	19
3	3	2
4	1	0
≥5	0	0
Other TKI	2	2
Number of regimens		
1	2	2
≥2	0	0
Immunotherapy	10	7
Number of regimens		

Subsequent therapy	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
number of patients		
1	9	7
2	1	0
≥3	0	0
VEGF inhibitor	16	12
Number of regimens		
1	13	12
2	3	0
≥3	0	0
Other	4	5
Number of regimens		
1	4	5
≥2	0	0
Non-Asian	**n=105**	**n=104**
Received any subsequent therapy	50	75
Number of regimens		
1	29	41
2	12	22
3	5	6
4	2	5
≥5	2	1
Cytotoxic chemotherapy	41	35
Number of regimens		
1	29	24
2	9	10
3	3	1
≥4	0	0
Platinum-containing chemotherapy	39	32
EGFR-TKI	20	60
Number of regimens		
1	16	49
2	3	7
3	1	3
4	0	1

Subsequent therapy	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
number of patients		
≥5	0	0
Other TKI	2	2
Number of regimens		
1	2	2
≥2	0	0
Immunotherapy	8	10
Number of regimens		
1	8	10
≥2	0	0
VEGF inhibitor	6	11
Number of regimens		
1	6	10
2	0	1
≥3	0	0
Other	1	2
Number of regimens		
1	1	2
≥2	0	0

B

Subsequent therapy	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
number of patients		
Ex19del	**n=175**	**n=174**
Received any subsequent therapy	79	120
Number of regimens		
1	38	63
2	21	31
3	9	14
4	7	9
≥5	4	3
Cytotoxic chemotherapy	65	58
Number of regimens		

Subsequent therapy	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
1	43	38
2	14	17
3	7	2
4	0	1
≥5	1	0
Platinum-containing chemotherapy	64	51
EGFR-TKI	39	101
Number of regimens		
1	32	79
2	4	17
3	2	4
4	1	1
≥5	0	0
Other TKI	1	3
Number of regimens		
1	1	3
≥2	0	0
Immunotherapy	11	10
Number of regimens		
1	10	10
2	1	0
≥3	0	0
VEGF inhibitor	12	15
Number of regimens		
1	10	14
2	2	1
≥3	0	0
Other	5	5
Number of regimens		
1	5	5
≥2	0	0
L858R	**n=104**	**n=103**
Received any subsequent therapy	54	60
Number of regimens		

Subsequent therapy	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
1	30	33
2	14	16
3	3	6
4	7	2
≥5	0	3
Cytotoxic chemotherapy	40	35
Number of regimens		
1	31	28
2	7	4
3	2	2
4	0	1
≥5	0	0
Platinum-containing chemotherapy	35	33
EGFR-TKI	30	41
Number of regimens		
1	23	31
2	5	9
3	2	1
≥4	0	0
Other TKI	3	1
Number of regimens		
1	3	1
≥2	0	0
Immunotherapy	7	7
Number of regimens		
1	7	7
≥2	0	0
VEGF inhibitor	10	8
Number of regimens		
1	9	8
2	1	0
≥3	0	0
Other	0	2
Number of regimens		

Subsequent therapy	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
1	0	2
≥2	0	0

Post-treatment anti-cancer therapies are those with a start date on or after the last dose date of randomized study treatment. Subjects with no post-treatment anti-cancer therapy have discontinued randomized study treatment without starting any other post-treatment. Subjects may be counted in multiple rows if they received more than one anti-cancer therapy or a combination therapy which contains drug substances from multiple classifications.

EGFR, epidermal growth factor receptor; Ex19del, Exon 19 deletion mutations; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor.

Table S5. Adverse events possibly causally-related to study treatment (investigator assessed) reported in at least 10% of the patients treated in either treatment arm

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
number of patients (percent)		
Diarrhea	140 (50)	144 (52)
Paronychia	84 (30)	77 (28)
Dry skin	86 (31)	86 (31)
Stomatitis	70 (25)	48 (17)
Dermatitis acneiform	69 (25)	132 (48)
Pruritus	44 (16)	38 (14)
Decreased appetite	36 (13)	35 (13)
Rash maculo-papular	34 (12)	44 (16)
Aspartate aminotransferase increased	23 (8)	58 (21)
Alanine aminotransferase increased	17 (6)	62 (22)

Table S6. Adverse events of grade 3 or higher by preferred term

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
number of patients (percent)		
Hyponatremia	9 (3)	4 (1)
Pneumonia	8 (3)	7 (3)
Diarrhea	7 (3)	8 (3)
Decreased appetite	7 (3)	5 (2)
Anemia	7 (3)	3 (1)
Ejection fraction decrease	5 (2)	1 (<1)
Renal*	5 (2)	1 (<1)
Neutrophil count decreased	5 (2)	1 (<1)
Prolonged QT interval on electrocardiogram	4 (1)	2 (1)
Sepsis	4 (1)	2 (1)
Pulmonary embolism	4 (1)	1 (<1)
Gamma-glutamyltransferase increased	4 (1)	0
Lymphocyte count decreased	4 (1)	0
Rashes and acnes*	3 (1)	20 (7)
Interstitial lung disease	3 (1)	3 (1)
Hypokalemia	3 (1)	3 (1)
Fatigue	3 (1)	2 (1)
Hypertension	3 (1)	1 (<1)
Cataract disorder	3 (1)	1 (<1)
Weight decreased	3 (1)	0
Alanine aminotransferase increased	2 (1)	25 (9)
Aspartate aminotransferase increased	2 (1)	12 (4)
Dyspnea	2 (1)	3 (1)
Nail effects*	2 (1)	2 (1)
Asthenia	2 (1)	2 (1)
Stomatitis	2 (1)	1 (<1)
Platelet count decreased	2 (1)	1 (<1)
Abdominal pain	2 (1)	0
Embolism	2 (1)	0
Headache	2 (1)	0
Urinary tract infection	2 (1)	0

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
Urticaria	2 (1)	0
White blood cell count decreased	2 (1)	0
Hemoptysis	1 (<1)	4 (1)
Dry skin*	1 (<1)	3 (1)
Abnormal hepatic function	1 (<1)	3 (1)
Pneumothorax	1 (<1)	2 (1)
Pruritus*	1 (<1)	0
Brain edema	1 (<1)	1 (<1)
Cerebral infarction	1 (<1)	1 (<1)
Cognitive disorder	1 (<1)	1 (<1)
Deep vein thrombosis	1 (<1)	1 (<1)
Gastroenteritis	1 (<1)	1 (<1)
Pleural effusion	1 (<1)	1 (<1)
Procedural pneumothorax	1 (<1)	1 (<1)
Urinary tract infection bacterial	1 (<1)	1 (<1)
Acute psychosis	1 (<1)	0
Acute myocardial infarction	1 (<1)	0
Anxiety	1 (<1)	0
Atrial fibrillation	1 (<1)	0
Abdominal pain upper	1 (<1)	0
Bladder cancer	1 (<1)	0
Benign prostatic hyperplasia	1 (<1)	0
Cerebral ischemia	1 (<1)	0
Clostridial infection	1 (<1)	0
Clostridium difficile colitis	1 (<1)	0
Cardiac arrest	1 (<1)	0
Cardiac failure chronic	1 (<1)	0
Dehydration	1 (<1)	0
Depression	1 (<1)	0
Depressed level of consciousness	1 (<1)	0
Desmoid cancer	1 (<1)	0
Diabetic ketoacidosis	1 (<1)	0
Dysphagia	1 (<1)	0

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
Endometrial cancer	1 (<1)	0
Fall	1 (<1)	0
General physical health deterioration	1 (<1)	0
Hepatitis	1 (<1)	0
Head injury	1 (<1)	0
Hemorrhoidal hemorrhage	1 (<1)	0
Hemorrhagic stroke	1 (<1)	0
Histiocytic necrotizing lymphadenitis	1 (<1)	0
Hyperamylasemia	1 (<1)	0
Hyperkalemia	1 (<1)	0
Hypermagnesemia	1 (<1)	0
Hypochloremia	1 (<1)	0
Hypoxia	1 (<1)	0
Inappropriate antidiuretic hormone secretion	1 (<1)	0
Intestinal ischemia	1 (<1)	0
Invasive ductal breast carcinoma	1 (<1)	0
Iron deficiency anemia	1 (<1)	0
Lower gastrointestinal hemorrhage	1 (<1)	0
Lower respiratory tract infection	1 (<1)	0
Lymphopenia	1 (<1)	0
Myocardial infarction	1 (<1)	0
Neutropenia	1 (<1)	0
Pancytopenia	1 (<1)	0
Pancreatitis	1 (<1)	0
Pneumonia mycoplasmal	1 (<1)	0
Pneumonia pseudomonal	1 (<1)	0
Post-procedural infection	1 (<1)	0
Presyncope	1 (<1)	0
Procedural pain	1 (<1)	0
Respiratory syncytial virus bronchitis	1 (<1)	0
Respiratory tract infection	1 (<1)	0
Retinal detachment	1 (<1)	0
Skin infection	1 (<1)	0

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
Skin toxicity	1 (<1)	0
Subdural hematoma	1 (<1)	0
Superior vena cava occlusion	1 (<1)	0
Syncope	1 (<1)	0
Thrombophlebitis	1 (<1)	0
Tracheobronchitis	1 (<1)	0
Transaminases increased	1 (<1)	0
Transient ischemic attack	1 (<1)	0
Type 2 diabetes mellitus	1 (<1)	0
Urethritis	1 (<1)	0
Venous thrombosis	1 (<1)	0
Vertigo positional	1 (<1)	0
Vomiting	0	4 (1)
Drug-induced liver injury	0	3 (1)
Blood alkaline phosphatase increased	0	3 (1)
Hypoalbuminemia	0	2 (1)
Hyperglycemia	0	2 (1)
Cystitis	0	2 (1)
Respiratory failure	0	2 (1)
Mouth ulceration	0	2 (1)
Pathological fracture	0	2 (1)
Sinusitis	0	1 (<1)
Hypocalcemia	0	1 (<1)
Device-related infection	0	1 (<1)
Rectal abscess	0	1 (<1)
Lower respiratory tract infection viral	0	1 (<1)
Perineal abscess	0	1 (<1)
Pharyngotonsillitis	0	1 (<1)
Endocarditis	0	1 (<1)
Clostridium difficile infection	0	1 (<1)
Bronchitis	0	1 (<1)
Bacteremia	0	1 (<1)
Anal abscess	0	1 (<1)

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
Endometrial adenocarcinoma	0	1 (<1)
Ovarian fibroma	0	1 (<1)
Prostate cancer	0	1 (<1)
Uterine leiomyoma	0	1 (<1)
Hypothyroidism	0	1 (<1)
Hypoglycemia	0	1 (<1)
Hypophosphatemia	0	1 (<1)
Restlessness	0	1 (<1)
Cerebrovascular accident	0	1 (<1)
Epilepsy	0	1 (<1)
Migraine	0	1 (<1)
Parkinsonism	0	1 (<1)
Peripheral motor neuropathy	0	1 (<1)
Seizure	0	1 (<1)
Somnolence	0	1 (<1)
Spinal cord compression	0	1 (<1)
Subarachnoid hemorrhage	0	1 (<1)
Blepharitis	0	1 (<1)
Vitreous detachment	0	1 (<1)
Vertigo	0	1 (<1)
Angina pectoris	0	1 (<1)
Bundle branch block left	0	1 (<1)
Pericardial effusion	0	1 (<1)
Pericarditis	0	1 (<1)
Ventricular extrasystoles	0	1 (<1)
Aortic dissection	0	1 (<1)
Circulatory collapse	0	1 (<1)
Peripheral artery occlusion	0	1 (<1)
Systolic hypertension	0	1 (<1)
Asthma	0	1 (<1)
Pneumonia aspiration	0	1 (<1)
Pneumonitis	0	1 (<1)
Pulmonary hemorrhage	0	1 (<1)

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
Gastrointestinal hemorrhage	0	1 (<1)
Melena	0	1 (<1)
Esophagitis	0	1 (<1)
Hepatic failure	0	1 (<1)
Skin irritation	0	1 (<1)
Toxic epidermal necrolysis	0	1 (<1)
Erythema nodosum	0	1 (<1)
Musculoskeletal chest pain	0	1 (<1)
Uterine polyp	0	1 (<1)
Condition aggravated	0	1 (<1)
Death	0	1 (<1)
Drug withdrawal syndrome	0	1 (<1)
Necrosis	0	1 (<1)
Pyrexia	0	1 (<1)
Blood bilirubin increased	0	1 (<1)
Protein urine present	0	1 (<1)
Femoral neck fracture	0	1 (<1)

*Grouped term.

Table S7. Serious adverse events by preferred term

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
number of patients (percent)		
Pneumonia	9 (3)	7 (3)
Interstitial lung disease	4 (1)	3 (1)
Pleural effusion	4 (1)	3 (1)
Hyponatremia	4 (1)	1 (<1)
Pulmonary embolism	4 (1)	1 (<1)
Diarrhea	3 (1)	4 (1)
Sepsis	3 (1)	3 (1)
Pyrexia	3 (1)	2 (1)
Asthenia	2 (1)	2 (1)
Decreased appetite	2 (1)	2 (1)
Dyspnea	2 (1)	2 (1)
Pneumothorax	2 (1)	2 (1)
Gastroenteritis	2 (1)	1 (<1)
Pneumonitis	2 (1)	1 (<1)
Abdominal pain	2 (1)	0
Enterocolitis	2 (1)	0
Acute kidney injury	2 (1)	0
Deep vein thrombosis	2 (1)	0
Vomiting	1 (<1)	5 (2)
Alanine aminotransferase increased	1 (<1)	2 (1)
Injury by fall	1 (<1)	2 (1)
Angina pectoris	1 (<1)	1 (<1)
Aspartate aminotransferase increased	1 (<1)	1 (<1)
Brain edema	1 (<1)	1 (<1)
Cerebral infarction	1 (<1)	1 (<1)
Clostridial difficile infection	1 (<1)	1 (<1)
Cognitive disorder	1 (<1)	1 (<1)
Confusional state	1 (<1)	1 (<1)
Fatigue	1 (<1)	1 (<1)
Platelet count decreased	1 (<1)	1 (<1)
Procedural pneumothorax	1 (<1)	1 (<1)

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
Abdominal pain upper	1 (<1)	0
Acute myocardial infarction	1 (<1)	0
Acute psychosis	1 (<1)	0
Atrial fibrillation	1 (<1)	0
Back pain	1 (<1)	0
Benign prostatic hyperplasia	1 (<1)	0
Bladder cancer	1 (<1)	0
Cardiac arrest	1 (<1)	0
Cardiac failure chronic	1 (<1)	0
Cardiac tamponade	1 (<1)	0
Cerebral ischemia	1 (<1)	0
Clostridial infection	1 (<1)	0
Dehydration	1 (<1)	0
Depressed level of consciousness	1 (<1)	0
Desmoid tumor	1 (<1)	0
Diabetic ketoacidosis	1 (<1)	0
Electrocardiogram QT prolonged	1 (<1)	0
Embolism	1 (<1)	0
Empyema	1 (<1)	0
Enteritis infectious	1 (<1)	0
Femur fracture	1 (<1)	0
Gastritis erosive	1 (<1)	0
Head injury	1 (<1)	0
Hemorrhagic stroke	1 (<1)	0
Hemorrhoidal hemorrhage	1 (<1)	0
Hepatic function abnormal	1 (<1)	0
Histiocytic necrotizing lymphadenitis	1 (<1)	0
Hypoxia	1 (<1)	0
Inappropriate antidiuretic hormone secretion	1 (<1)	0
Inguinal hernia	1 (<1)	0
Intestinal ischemia	1 (<1)	0
Invasive ductal breast carcinoma	1 (<1)	0

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
Lower gastrointestinal hemorrhage	1 (<1)	0
Lower respiratory tract infection	1 (<1)	0
Myocardial infarction	1 (<1)	0
Neck pain	1 (<1)	0
Pancreatitis	1 (<1)	0
Pancytopenia	1 (<1)	0
Pneumonia mycoplasmal	1 (<1)	0
Pneumonia pseudomal	1 (<1)	0
Pneumonia viral	1 (<1)	0
Post procedural infection	1 (<1)	0
Renal failure	1 (<1)	0
Respiratory syncytial virus bronchitis	1 (<1)	0
Respiratory tract infection	1 (<1)	0
Rotator cuff syndrome	1 (<1)	0
Spinal cord compression	1 (<1)	0
Stomatitis	1 (<1)	0
Subdural hematoma	1 (<1)	0
Tachyarrhythmia	1 (<1)	0
Thrombophlebitis	1 (<1)	0
Tonsillolith	1 (<1)	0
Tracheobronchitis	1 (<1)	0
Transaminases increased	1 (<1)	0
Transient ischemic attack	1 (<1)	0
Urinary tract infection	1 (<1)	0
Venous thrombosis	1 (<1)	0
Wound complication	1 (<1)	0
Wrist fracture	1 (<1)	0
Hemoptysis	0	4 (1)
Drug-induced liver injury	0	3 (1)
Cystitis	0	2 (1)
Dermatitis acneiform	0	2 (1)
Mouth ulceration	0	2 (1)
Respiratory failure	0	2 (1)

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
Anemia	0	1 (<1)
Anal abscess	0	1 (<1)
Aortic dissection	0	1 (<1)
Asthma	0	1 (<1)
Bacteremia	0	1 (<1)
Bronchitis	0	1 (<1)
Bundle branch block left	0	1 (<1)
Cerebrovascular accident	0	1 (<1)
Chest pain	0	1 (<1)
Chronic gastritis	0	1 (<1)
Circulatory collapse	0	1 (<1)
Condition aggravated	0	1 (<1)
Death	0	1 (<1)
Device related infection	0	1 (<1)
Diarrhea infectious	0	1 (<1)
Diverticulitis	0	1 (<1)
Dizziness	0	1 (<1)
Drug withdrawal syndrome	0	1 (<1)
Endocarditis	0	1 (<1)
Endometrial adenocarcinoma	0	1 (<1)
Epilepsy	0	1 (<1)
Erythema nodosum	0	1 (<1)
Escherichia urinary tract infection	0	1 (<1)
Febrile infection	0	1 (<1)
Femoral neck fracture	0	1 (<1)
Flank pain	0	1 (<1)
Gastrointestinal hemorrhage	0	1 (<1)
General physical health deterioration	0	1 (<1)
Hyperglycemia	0	1 (<1)
Hypoglycemia	0	1 (<1)
Hypothyroidism	0	1 (<1)
Ischemic stroke	0	1 (<1)
Lower respiratory tract infection viral	0	1 (<1)

Adverse event	Osimertinib (n=279)	Comparator EGFR-TKI (n=277)
Melaena	0	1 (<1)
Migraine	0	1 (<1)
Musculoskeletal chest pain	0	1 (<1)
Esophagitis	0	1 (<1)
Orthostatic hypotension	0	1 (<1)
Ovarian fibroma	0	1 (<1)
Pathological fracture	0	1 (<1)
Pericarditis	0	1 (<1)
Perineal abscess	0	1 (<1)
Peripheral artery occlusion	0	1 (<1)
Peripheral motor neuropathy	0	1 (<1)
Pharyngotonsillitis	0	1 (<1)
Pleural infection	0	1 (<1)
Prostate cancer	0	1 (<1)
Pulmonary hemorrhage	0	1 (<1)
Rectal abscess	0	1 (<1)
Salmonella bacteremia	0	1 (<1)
Seizure	0	1 (<1)
Sinusitis	0	1 (<1)
Subarachnoid hemorrhage	0	1 (<1)
Toxic epidermal necrolysis	0	1 (<1)
Ureterolithiasis	0	1 (<1)
Uterine leiomyoma	0	1 (<1)
Uterine polyp	0	1 (<1)
Vertigo	0	1 (<1)

REFERENCES

1. Berry G, Kitchin RM, Mock PA. A comparison of two simple hazard ratio estimators based on the logrank test. Statistics in medicine 1991;10:749-55.
2. Selke TaS, D. Sequential Analysis of the Proportional Hazards Model. Biometrika 1983;70:315-26.
3. Gail M, Simon R. Testing for qualitative interactions between treatment effects and patient subsets. Biometrics 1985;41:361-72.